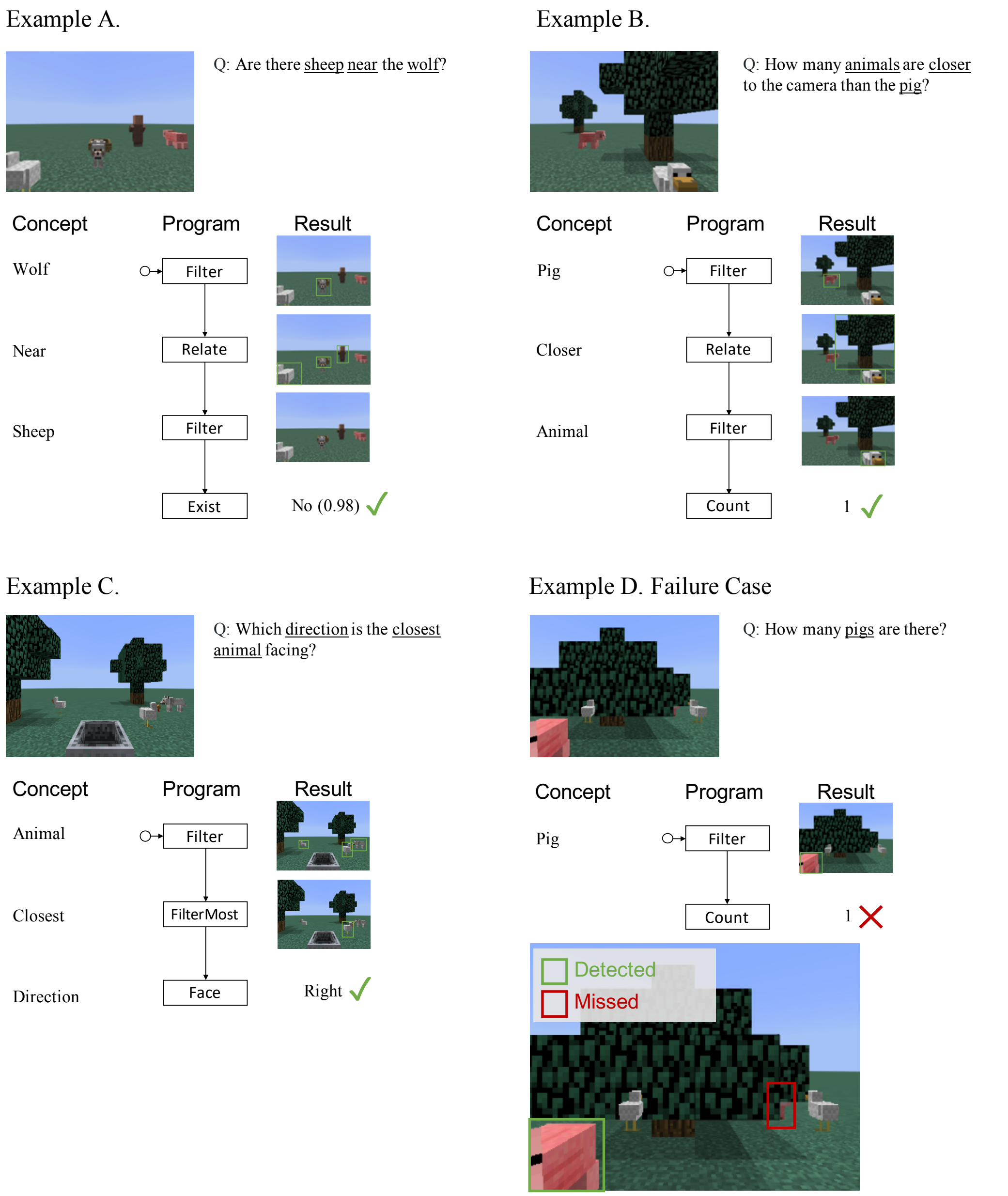

Figure 12: Exemplar execution trace generated by our Neuro-Symbolic Concept Learner on the Minecraft reasoning dataset. Example A, B and C are successful execution. Example C demonstrates the semantics of the FilterMost operation. Example D shows a failure case: the detection model fails to detect a pig hiding behind the big tree.

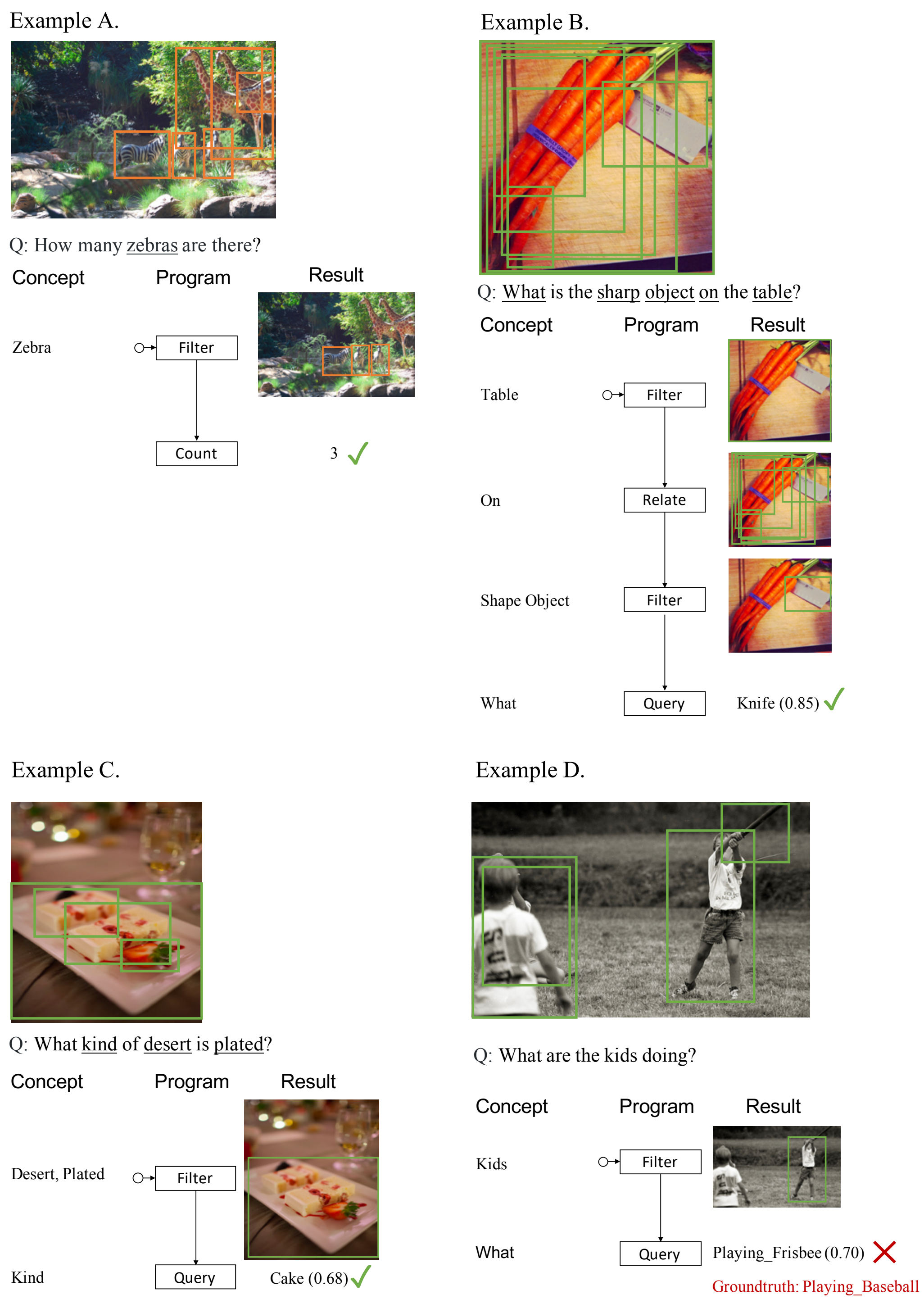

Figure 13: Illustrative execution trace generated by our Neuro-Symbolic Concept Learner on the VQS dataset. Execution traces A and B shown in the figure leads to the correct answer to the question. Our model effectively learns visual concepts from data. The symbolic reasoning process brings transparent execution trace and can easily handle quantities (e.g., object counting in Example A). In Example C, although NS-CL answers the question correctly, it locates the wrong object during reasoning: a dish instead of the cake. In Example D, our model misclassifies the sport as frisbee.

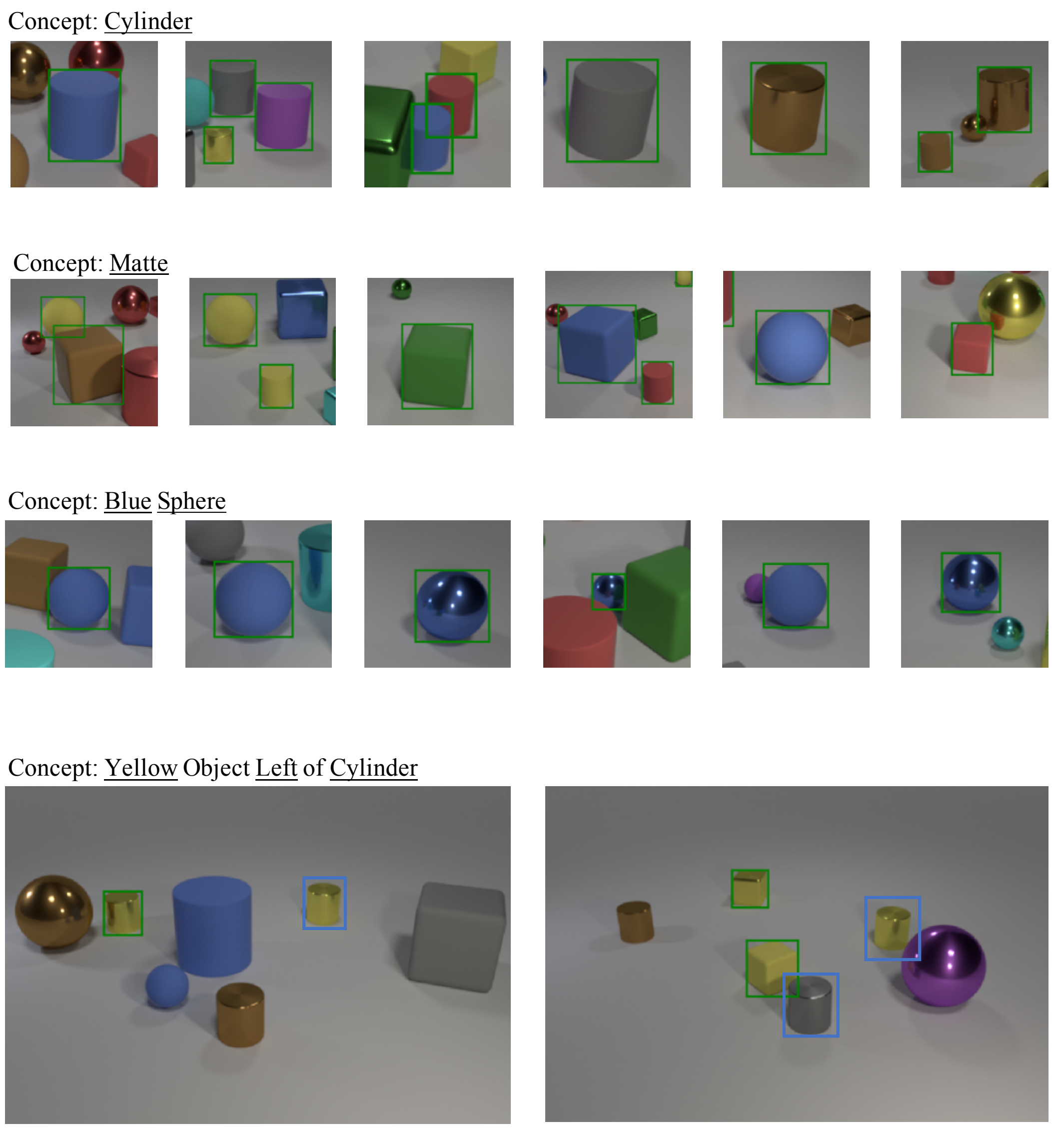

Figure 14: Concepts learned on the CLEVR dataset.